

1.3 The Juggler—Multitasking

Before we move further, there is the small matter of Processes and Threads to be dealt with.

1.3.1 Processes And Threads

A *process* is the running version of a computer program. It is given its own private space in the system memory which it uses to store its code and data. The *context* of the process is the entire contents of this memory, and some more data about the system's state. This way, when the context of a process is loaded into the memory, it can start executing from when it left off. Processes are sometimes called “virtual machines”, because each process is given the illusion that it is the only one running on the system. Thus, many virtual machines run on the same physical machine, all managed by the operating system.

A *thread* is a sub-process created by a “parent” process. It runs in the same memory area as the process that creates it, and shares data with other threads created by that process. One process may create many threads to do different things at the same time. For example, while one thread of Microsoft Word brings your input onto the screen, another thread checks your spelling as you type, and yet another is saving backups of your document—just in case.

1.3.2 From Multiprogramming To Multithreading

Multitasking, simply put, is the ability of an operating system to do many different things at the same time. It sounds simple enough, but how do you do many different things at the same time when all you have at your disposal is a processor that can only process *one* instruction at a time?

Seeing how many threads are created by a Process

In Windows XP, hit [Ctrl] + [Shift] + [Esc] to bring up the Task Manager. Under the Performance tab, you will see “Threads” and “Processes” in the box labelled “Totals”. Now go to your Start Menu and select a program (let's say Microsoft Word). You will notice that while only one process has been added, the number of threads has increased by four or five.]